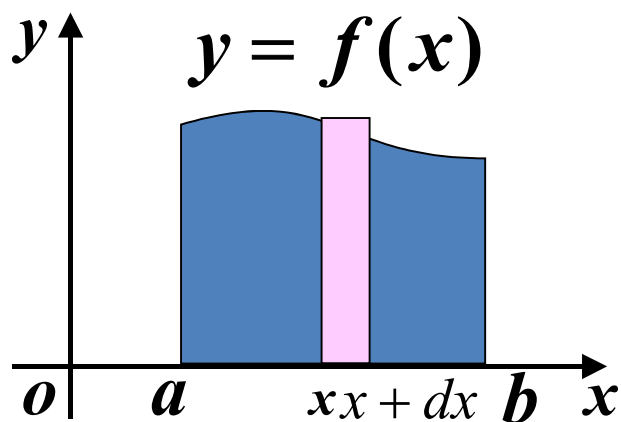


§ 7.2 几何应用

- 平面图形的面积
- 平面曲线的弧长
- 立体体积

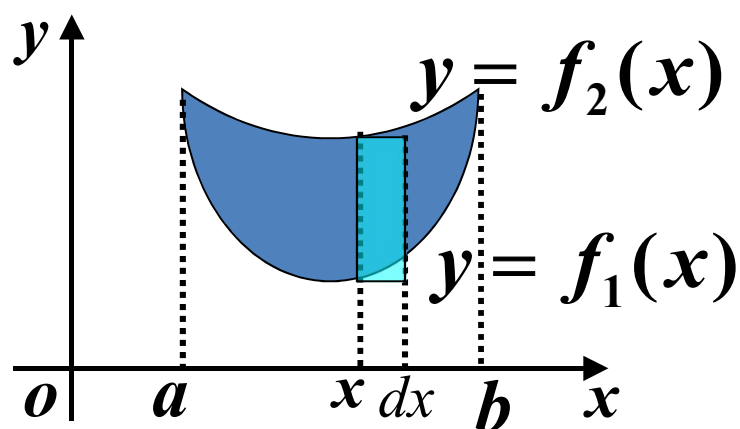
一、平面图形的面积

1、直角坐标系情形 “X”型区域



曲边梯形的面积

$$A = \int_a^b f(x) dx$$

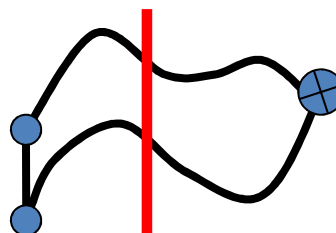
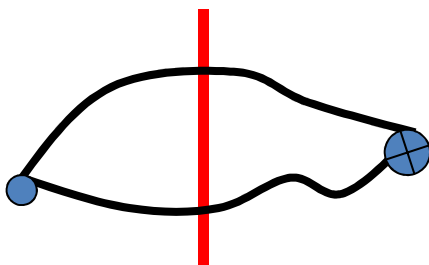
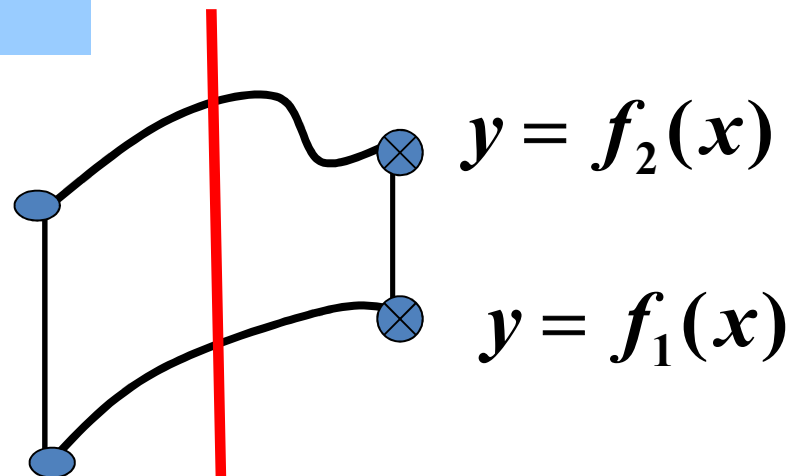
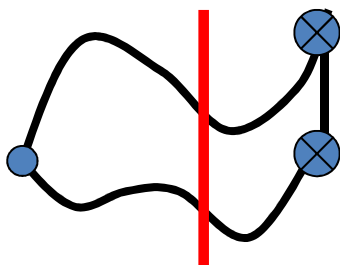


曲边梯形的面积

$$A = \int_a^b [f_2(x) - f_1(x)] dx$$

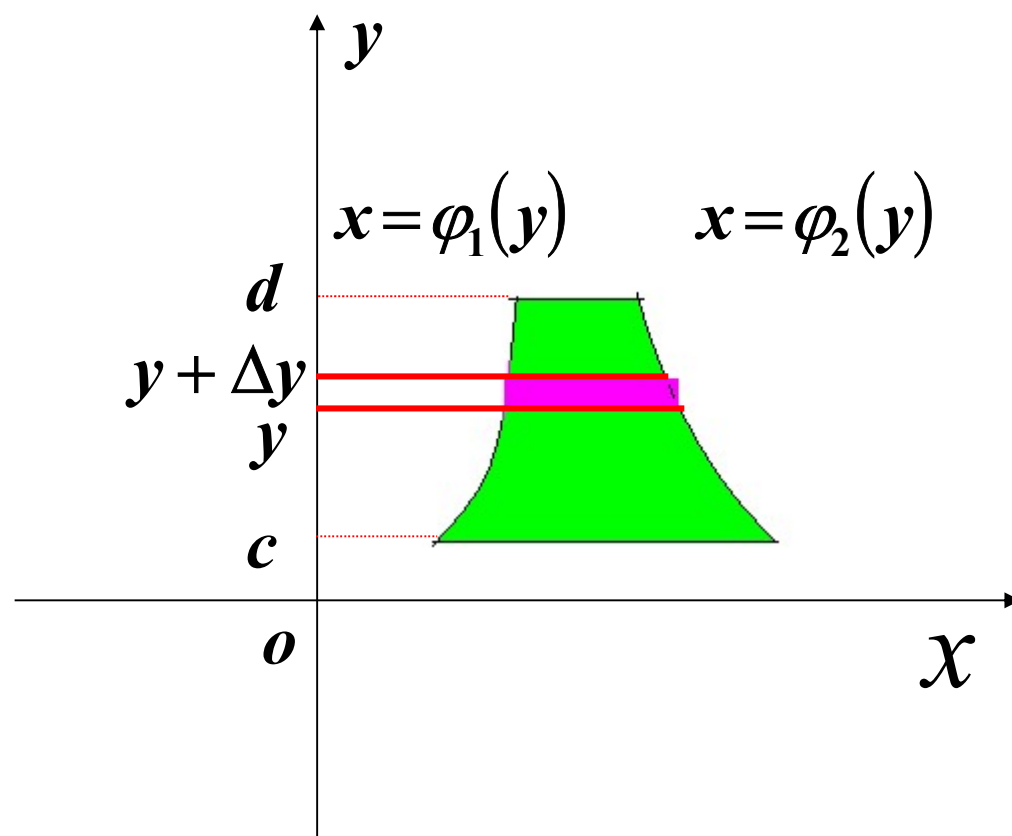
上面求平面图形的面积公式中平面图形向 x 轴投影, x 是积分变量.

“X”型区域:



若平面图形向 y 轴投影, y 是积分变量,
则有求面积公式

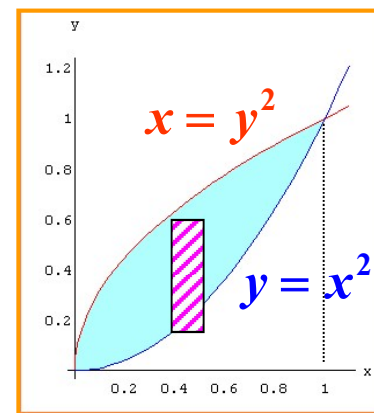
$$A = \int_c^d [\varphi_2(y) - \varphi_1(y)] dy$$



例 1 计算由两条抛物线 $y^2 = x$ 和 $y = x^2$ 所围成的图形的面积.

解 两曲线的交点
 $(0,0)$ $(1,1)$

法一: 选 x 为积分变量 $x \in [0,1]$



面积元素 $dA = (\sqrt{x} - x^2)dx$

$$A = \int_0^1 (\sqrt{x} - x^2)dx = \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{x^3}{3} \right]_0^1 = \frac{1}{3}.$$

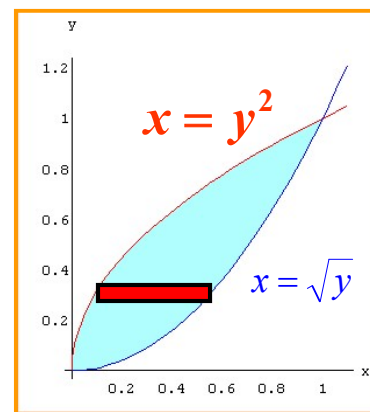
法二： 选 y 为积分变量 $y \in [0,1]$

面积元素

$$dA = (\sqrt{y} - y^2)dy$$

$$A = \int_0^1 (\sqrt{y} - y^2)dy$$

$$\begin{aligned} &= \left[\frac{2}{3} y^{\frac{3}{2}} - \frac{y^3}{3} \right]_0^1 \\ &= \frac{1}{3}. \end{aligned}$$



例 2 计算由曲线 $y = x^3 - 6x$ 和 $y = x^2$ 所围成的图形的面积.

解 两曲线的交点

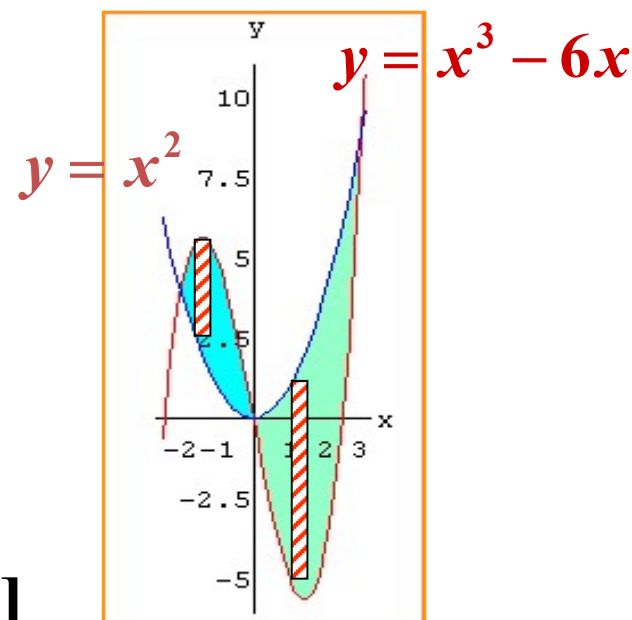
$$\begin{cases} y = x^3 - 6x \\ y = x^2 \end{cases}$$

$$\Rightarrow (0,0), (-2,4), (3,9).$$

选 x 为积分变量 $x \in [-2, 3]$

$$(1) \quad x \in [-2, 0], \quad dA_1 = (x^3 - 6x - x^2)dx$$

$$(2) \quad x \in [0, 3], \quad dA_2 = (x^2 - x^3 + 6x)dx$$



于是所求面积 $A = A_1 + A_2$

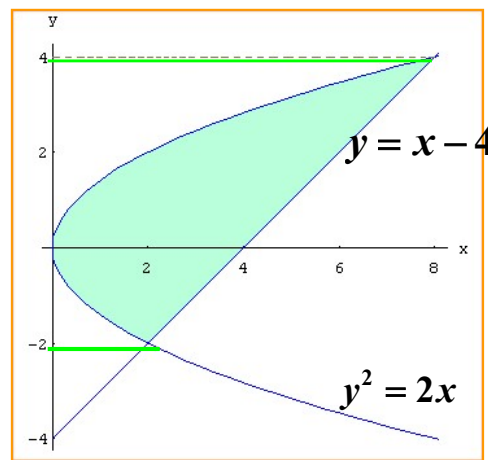
$$\begin{aligned} A &= \int_{-2}^0 (x^3 - 6x - x^2) dx + \int_0^3 (x^2 - x^3 + 6x) dx \\ &= \frac{253}{12}. \end{aligned}$$

说明：注意各积分区间上被积函数的形式.

例 3 计算由曲线 $y^2 = 2x$ 和直线 $y = x - 4$ 所围成的图形的面积.

解 两曲线的交点

$$\begin{cases} y^2 = 2x \\ y = x - 4 \end{cases} \Rightarrow (2, -2), (8, 4).$$



法一: 选 y 为积分变量 $y \in [-2, 4]$

$$dA = \left(y + 4 - \frac{y^2}{2} \right) dy \quad A = \int_{-2}^4 \left(y + 4 - \frac{y^2}{2} \right) dy = 18.$$

法二： 选 x 为积分变量

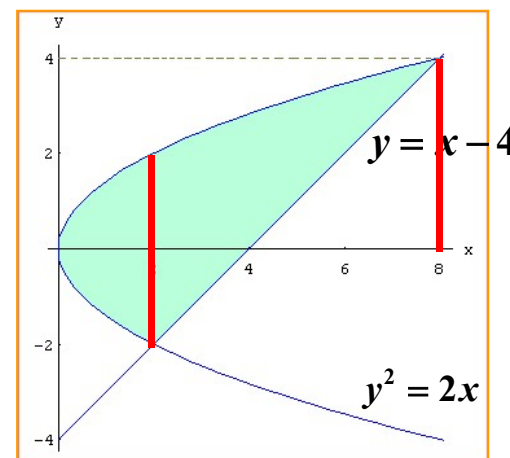
$$x \in [0, 8] = [0, 2] \cup [2, 8]$$

$$(1) \quad x \in [0, 2], \quad dA_1 = \left[\sqrt{2x} - (-\sqrt{2x}) \right] dx$$

$$(2) \quad x \in [2, 8], \quad dA_2 = \left[\sqrt{2x} - (x - 4) \right] dx$$

于是所求面积 $A = A_1 + A_2$

$$\begin{aligned} A &= \int_0^2 (2\sqrt{2x}) dx + \int_2^8 (\sqrt{2x} - x + 4) dx \\ &= 18. \end{aligned}$$



如果曲边梯形的曲边为参数方程 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$

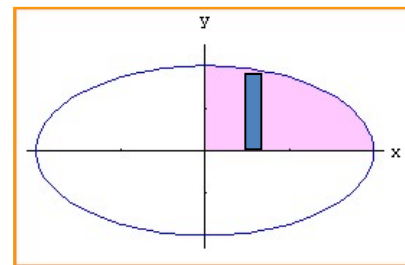
曲边梯形的面积 $A = \int_{t_1}^{t_2} |\psi(t)| \phi'(t) dt.$

(其中 t_1 和 t_2 对应曲线起点与终点的参数值)

在 $[t_1, t_2]$ (或 $[t_2, t_1]$) 上 $x = \varphi(t)$ 具有连续导数,
 $y = \psi(t)$ 连续.

例 4 求椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 的面积.

解 椭圆的参数方程 $\begin{cases} x = a \cos t \\ y = b \sin t \end{cases}$



由对称性知总面积等于**4**倍第一象限部分面积.

$$A = 4 \int_0^a y dx = 4 \int_{\frac{\pi}{2}}^0 b \sin t d(a \cos t)$$

$$= 4ab \int_0^{\frac{\pi}{2}} \sin^2 t dt = \pi ab.$$

例 设 M 为曲线 $\begin{cases} x = \cos t \\ y = 2\sin^2 t \end{cases}, 0 \leq t \leq \frac{\pi}{2}$ 上的一点,

求此曲线与直线 OM 及 x 轴所界图形的面积 S ,

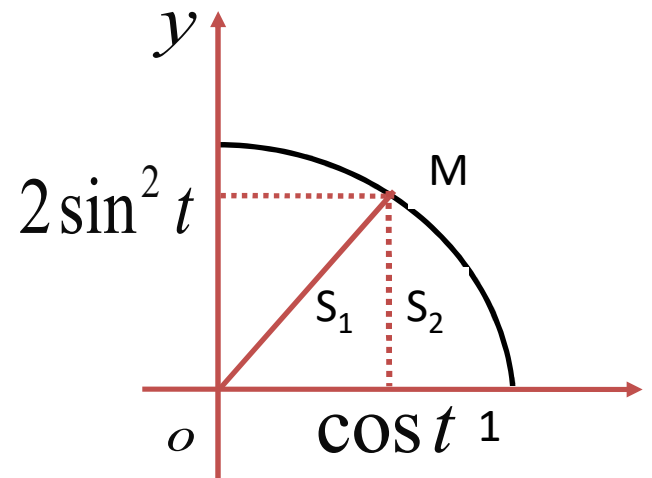
求 $\frac{dS}{dt}$ 取得最大值时, 点 M 的坐标

解 $S = S_1 + S_2$

$$= \frac{1}{2} \cos t (2\sin^2 t) + \int_{\cos t}^1 y dx$$

由于 $y = 2(1 - x^2)$

$$\Rightarrow S = \cos t \sin^2 t + \int_{\cos t}^1 2(1 - x^2) dx$$



$$\begin{aligned}\frac{dS}{dt} &= 2\sin t \cos^2 t - \sin^3 t + 2(1 - \cos^2 t)\sin t \\ &= 2\sin t - \sin^3 t\end{aligned}$$

$$\text{令 } \frac{d^2 S}{dt^2} = 2\cos t - 3\sin^2 t \cos t = 0$$

$$\Rightarrow t_1 = \frac{\pi}{2}, \quad t_2 = \arcsin \sqrt{\frac{2}{3}}$$

$$\text{由于 } \left. \frac{dS}{dt} \right|_{t=0} = 0, \quad \left. \frac{dS}{dt} \right|_{t=\frac{\pi}{2}} = 1, \quad \left. \frac{dS}{dt} \right|_{t=t_2} = \frac{4}{3} \sqrt{\frac{2}{3}} > 1$$

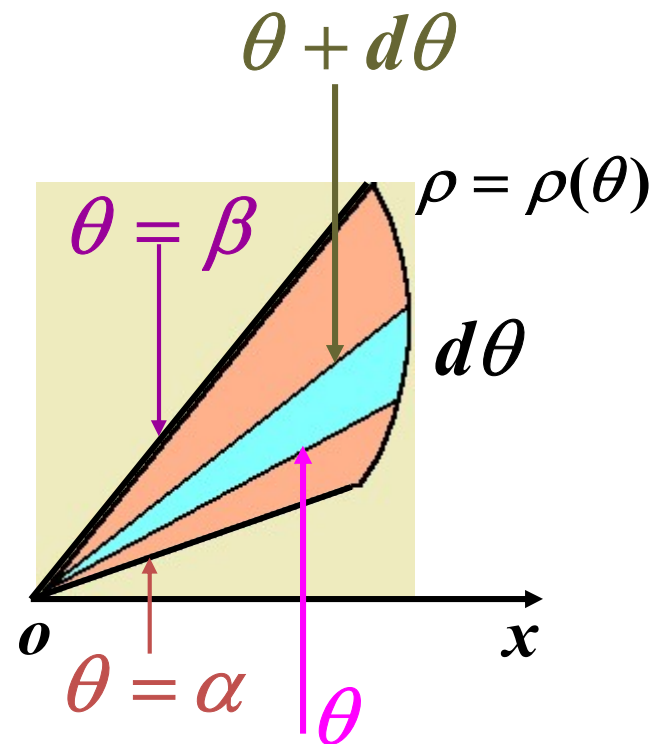
$$\therefore \text{当 } t = \arcsin \sqrt{\frac{2}{3}} \text{ 时, } \frac{dS}{dt} \text{ 取最大值,}$$

2、极坐标系情形

设由曲线 $\rho = \rho(\theta)$ 及射线 $\theta = \alpha$ 、 $\theta = \beta$ 围成一曲边扇形，求其面积．这里， $\rho(\theta)$ 在 $[\alpha, \beta]$ 上连续，且 $\rho(\theta) \geq 0$ ．

面积微元 $dA = \frac{1}{2}[\rho(\theta)]^2 d\theta$

曲边扇形的面积 $A = \int_{\alpha}^{\beta} \frac{1}{2}[\rho(\theta)]^2 d\theta$ ．

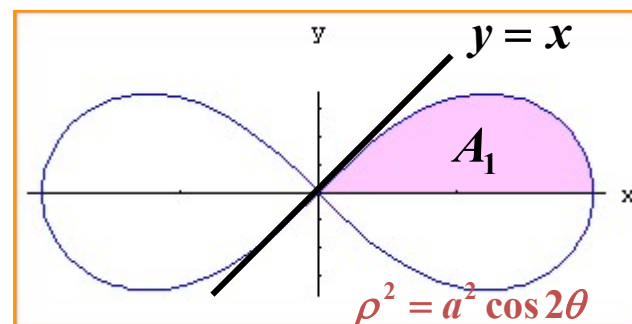


例 5 求双纽线 $\rho^2 = a^2 \cos 2\theta$ 所围平面图形的面积.

解 由对称性知总面积=4倍第一象限部分面积

$$A = 4A_1$$

$$A = 4 \int_0^{\frac{\pi}{4}} \frac{1}{2} a^2 \cos 2\theta d\theta = a^2.$$



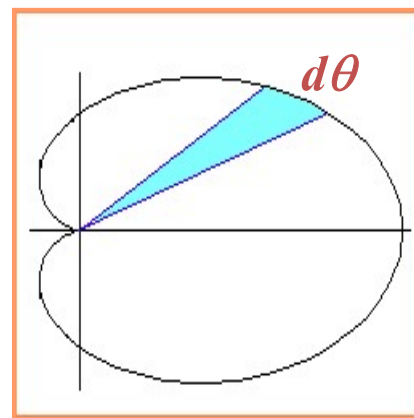
例 6 求心形线 $\rho = a(1 + \cos \theta)$ 所围平面图形的面积($a > 0$).

解 $dA = \frac{1}{2}a^2(1 + \cos \theta)^2 d\theta$

利用对称性知

$$A = 2 \cdot \frac{1}{2}a^2 \int_0^\pi (1 + \cos \theta)^2 d\theta$$
$$= a^2 \int_0^\pi (1 + 2\cos \theta + \cos^2 \theta) d\theta$$

$$= a^2 \left[\frac{3}{2}\theta + 2\sin \theta + \frac{1}{4}\sin 2\theta \right]_0^\pi = \frac{3}{2}\pi a^2.$$



例 求平面区域 $D = \left\{ (r, \theta) \left| \frac{1}{2} \leq r \leq 1 + \cos \theta \right. \right\}$ 的面积

解 画出草图, 如图所示

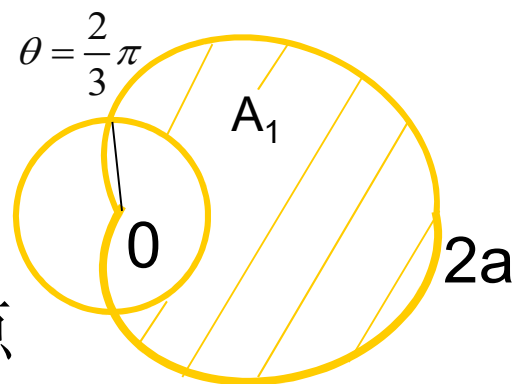
图形关于极轴 r 对称 $\Rightarrow A = 2A_1$

求 $r = \frac{1}{2}$ 与 $r = 1 + \cos \theta$ 的交点

解
$$\begin{cases} r = \frac{1}{2} \\ r = 1 + \cos \theta \end{cases} \quad \text{得} \quad \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \theta = \frac{2}{3}\pi \quad (0 \leq \theta \leq \pi)$$

$$\Rightarrow \text{积分区间} [0, \frac{2}{3}\pi]$$



$$A = 2A_1$$

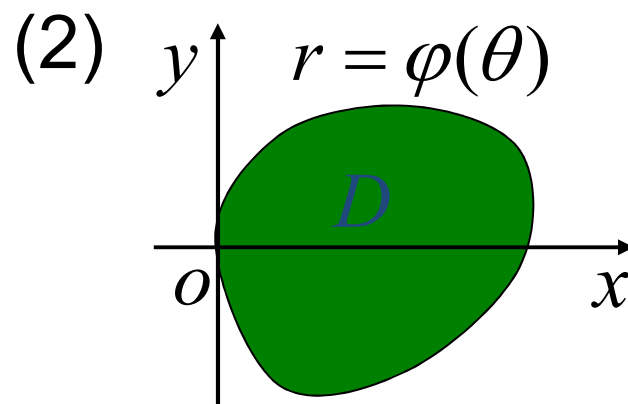
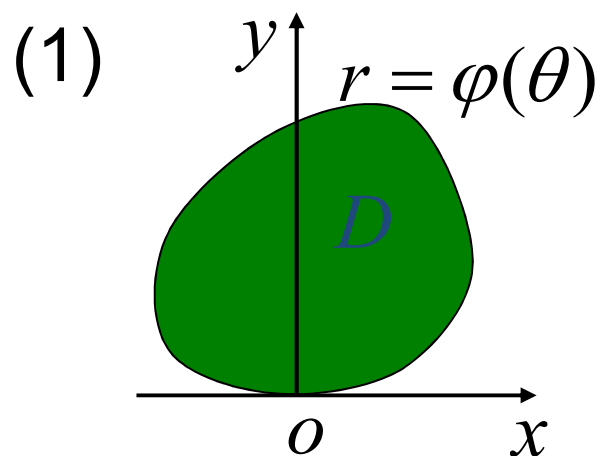
$$= 2 \int_0^{\frac{2}{3}\pi} \left[\frac{1}{2} (1 + \cos \theta)^2 - \frac{1}{2} \left(\frac{1}{2} \right)^2 \right] d\theta$$

$$= \int_0^{\frac{2}{3}\pi} \left(1 + 2 \cos \theta + \cos^2 \theta - \frac{1}{4} \right) d\theta$$

$$= \frac{5}{6} \pi + \frac{7}{8} \sqrt{3}$$



思考: 下列各图中域 D 分别与 x, y 轴相切于原点, 试问 θ 的变化范围是什么?



答: (1) $0 \leq \theta \leq \pi$; (2) $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$